Rhodoxy 50WSP Powder for Oral Solution 50%w/w

Composition :
Doxycycline hyclate 57.7%w/w
(equivalent to Doxycycline 50%w/w)

Product Description : Light yellow or yellow fine powder

Pharmacodynamics :
Doxycycline is a semisynthetic tetracycline derivative. It acts by inhibiting protein synthesis at the ribosomal level, predominantly by binding to the 30S ribosomal subunits of bacteria. Doxycycline is a broad-spectrum antibiotic. It exhibits a wide range of activity against Gram-positive and Gram-negative, aerobic and anaerobic pathogens, especially against *Pasteurella multocida* and *Mycoplasma hyopneumoniae* isolated from pig respiratory infections and *Mycoplasma gallisepticum* associated with clinical respiratory infections in chickens and turkeys.

Four resistance mechanisms acquired by micro-organisms against tetracyclines in general have been reported: decreased accumulation of tetracyclines (decreased permeability of the bacterial cell wall and active efflux), protein protection of the bacterial ribosome, enzymatic inactivation of the antibiotic and rRNA mutations (preventing the tetracycline binding to ribosome). Tetracycline resistance is usually acquired by means of plasmids or other mobile elements (e.g. conjugative transposons). Cross-resistance between tetracyclineis common but depends on the mechanism conferring resistance. Due to the greater liposolubility and greater ability to pass through cell membranes (in comparison to tetracycline), doxycycline retains a certain degree of efficacy against micro-organisms with acquired resistance to tetracyclines via efflux pumps. However, resistance mediated by ribosomal protection proteins confer cross-resistance to doxycycline.

Pharmacokinetics :
In general, doxycycline is quite rapidly and extensively absorbed from the gastrointestinal tract, widely distributed in the organism, not metabolised to any significant extent and excreted primarily in faeces, mostly in a microbiologically inactive form.

After oral administration to pigs, doxycycline is substantially absorbed from the gastrointestinal tract. The binding rate to plasma proteins is 93%. It is widely distributed in the organism; at the steady state, the volume of distribution (V_{ss}) is 1.2 L/kg. The elimination half-life was reported to be 4-4.2 hours in pigs. The steady state plasma concentrations of doxycycline after repeated oral administrations of the veterinary medicinal product at a dose of 20 mg/kg body weight for 5 days ranged from 1.0 and 1.5 µg/ml. Both the lung and nasal mucosa concentrations at steady-state were higher than the plasma level. The ratio between tissue and plasma concentration was found to be 1.3 for lung and 3.4 for nasal mucosa. The doxycycline concentrations both in the lung and the nasal mucosa exceeded the MIC_{90} of the drug against the target respiratory pathogens.

Pharmacokinetics of doxycycline after single oral administration to chickens and turkeys is characterized by a quite rapid and substantial absorption from the gastrointestinal tract providing peak plasma concentrations between 0.4 and 3.3 hours in chickens and 1.5 to 7.5 hours in turkeys, depending on age and the presence of food. The drug is widely distributed in the organism with V_d values close to or greater than 1, and exhibits a shorter elimination half-life in chickens (4.8 to 9.4 hours) than in turkeys (7.9 to 10.8 hours). The protein binding ratio at therapeutic plasma concentrations is in the range of 70-85%. The bioavailability in chickens and turkeys may vary between 41 and 73%, and 25 and 64%, respectively also depending on the age and feeding. The presence of food in the gastrointestinal tract determines a lower bioavailability compared to that obtained in the fasted state.

After continuous in-water administration of the veterinary medicinal product at doses of 20 mg doxycycline/kg (chickens) and 25 mg doxycycline/kg (turkeys) for 5 days the average plasma concentrations over the whole treatment period were reported as 1.86 ± 0.71 µg/ml in chickens and 2.24 ± 1.02 µg/ml in turkeys. In both avian species the PK/PD analysis of $f\mathrm{AUC}/\mathrm{MIC}_{90}$ data resulted in > 24h values that meet the requirements for tetracyclines.

Indication :
Pigs: treatment of clinical respiratory infections caused by *Mycoplasma hyopneumoniae* and *Pasteurella multocida* susceptible to doxycycline.

Chickens and turkeys: treatment of clinical respiratory infections associated with *Mycoplasma gallisepticum* susceptible to doxycycline.

Dosage and Administration:
In drinking water use.

Dosage:
In pigs and chickens:
20 mg doxycycline per kg body weight daily (equivalent to 40 mg product per kg body weight), administered in the drinking water for 5 consecutive days.

In turkeys:
25 mg doxycycline per kg body weight daily (equivalent to 50 mg product per kg body weight), administered in the drinking water for 5 consecutive days.

Administration:
Based on the recommended dosage and the number and weight of the animals to be treated, the exact daily amount of the product to be administered should be calculated according to the following formula:

$$\frac{\text{mg product per kg body weight per day} \times \text{mean body weight (kg) of animals to be treated}}{\text{Mean daily water consumption (litre per animal)}} = \text{mg product per litre of drinking water}$$

To ensure a correct dosage, body weight should be determined as accurately as possible to avoid underdosing.
The uptake of medicated water is dependent on the clinical condition of the animals. In order to obtain the correct dosage, the concentration in drinking water may have to be adjusted. The use of suitably calibrated weighing equipment is recommended if part packs are used. The daily amount is to be added to the drinking water in such a way that all medication will be consumed within 24 hours. Medicated drinking water should be freshly prepared every 24 hours. It is recommended to prepare a concentrated pre-

solution and to dilute this further to therapeutic concentrations, if required. Alternatively, the concentrated solution can be used in proportional water medicator. The maximum solubility of the product in water is at least 100 g/L.

It should be ensured that all animals intended to be treated should have free access to the drinking facilities. At the end of treatment, the watering equipment should be cleaned adequately to avoid the uptake of remaining quantities in sub-therapeutic doses. The medicated water must not be made or stored in a metal container or used in oxidized drinking equipment. Solubility of the product is pH-dependent and it may precipitate if it is mixed in hard alkaline drinking water.

Contraindication:
Do not use in case of known hypersensitivity to tetracycline or to the excipient.
Do not use when tetracycline resistance has been detected in the herd/flock due to the potential for cross-resistance.
Do not use in animals with impaired liver or kidney function.

Special warnings for each target species:
The uptake of medication by animals can be altered as a consequence of illness. In case of insufficient uptake of drinking water, animals should be treated parenterally.

Warning & Precaution:
Special precautions for use in animals
The safety of the product has not been established in piglets before weaning.
Inappropriate use of the product may increase the prevalence of bacteria resistant to tetracyclines due to the potential for cross-resistance.
Due to variability (time, geographical) in susceptibility of bacteria for doxycycline, bacteriological sampling and susceptibility testing of micro-organisms from diseased animals on farm are highly recommended.

Use of the product should be based on identification and susceptibility testing of the target pathogen(s). If this is not possible, therapy should be based on epidemiological information and knowledge of susceptibility of the target bacteria at farm level, or at local/regional level.
Use of the product should be in accordance with official, national and regional antimicrobial policies. Avoid administration in oxidized drinking equipment.
As eradication of the target pathogens may not be achieved, medication should therefore be combined with good management practices, e.g. good hygiene, proper ventilation, no overstocking.

Special precautions to be taken by the person administering the medicinal product to animals

- This product may cause contact dermatitis and/or hypersensitivity reactions if contact is made with the skin or eyes (powder and solution), or if the powder is inhaled.
- Take measures to avoid producing dust when incorporating the product into water. Avoid direct contact with skin and eyes when handling the product to prevent sensitisation and contact dermatitis.
- People with known hypersensitivity to tetracyclines should avoid contact with the veterinary medicinal product. During preparation and administration of the medicated drinking water, skin contact with the product and inhalation of dust particles should be avoided. Wear impermeable gloves (e.g. rubber or latex) and an appropriate dust mask (e.g. disposable half-mask respirator conforming to European Standard EN 149 or a non-disposable respirator to European Standard EN 140 with a filter to EN 143) when applying the product.

- In the event of eye or skin contact, rinse the affected area with large amounts of clean water and if irritation occurs, seek medical attention.
- Wash hands and contaminated skin immediately after handling the product.
- Do not smoke, eat or drink while handling the product.

If you develop symptoms following exposure such as skin rash, you should seek medical advice and show this warning to the physician. Swelling of the face, lips or eyes, or difficulty with breathing are more serious symptoms and require urgent medical attention.

Special precautions for the disposal of unused veterinary medicinal product or waste materials derived from the use of such products
Drug containers and any residual contents should be disposed of safely, according to the regulation in force.

Interaction with other medicaments:
Do not administer concurrently with feed overloaded with polyvalent cations such as Ca^{2+}, Mg^{2+}, Zn^{2+} and Fe^{3+} because the formation of doxycycline complexes with these cations is possible. It is advised that the interval between administration of other products containing polyvalent cations should be 1-2 hours because they limit the absorption of tetracycline.
Doxycycline has a low affinity for forming complexes with calcium and studies have demonstrated that doxycycline scarcely affects skeleton formation.
Do not administer together with antacids, kaolin or iron preparations.
Do not administer in conjunction with bactericidal antibiotics such as beta-lactames as tetracyclines are bacteriostatic antimicrobials.
Doxycycline increases the action of anticoagulants.

Statement on usage during pregnancy & lactation:
<u>Pregnancy and lactation</u>:
Laboratory studies in rats and rabbits have not produced any evidence of teratogenic, foetotoxic or maternotoxic effects.
The safety of the product has not been established in pregnant or lactating sows. Use is not recommended during pregnancy or lactation.

<u>Laying birds</u>:
Do not use in birds in lay or within 4 weeks before the onset of the laying period.

Adverse Effect:
As for all tetracyclines, on rare occasions allergic reactions and photosensitivity may occur. If suspected adverse reactions occur, treatment should be discontinued.

The frequency of adverse reactions is defined using the following convention:
- Very common (more than 1 in 10 animals treated displaying adverse reaction(s))
- Common (more than 1 but less than 10 animals in 100 animals treated)
- Uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- Rare (more than 1 but less than 10 animals in 10,000 animals treated)
- Very rare (less than 1 animal in 10,000 animals treated, including isolated reports)

Overdose and Treatment:
During the target animal tolerance study, no adverse effects were observed in any of the target animal species, even at the fivefold therapeutic dose administered for two times the recommended duration.

If suspected toxic reactions do occur due to extreme overdose, the medication should be discontinued and appropriate symptomatic treatment should be initiated, if necessary.

Storage Condition:
Store in dry place below 30°C. Protect from direct sunlight.

Withdrawal Period:
Meat and offal:
Pigs: 4 days
Chickens: 5 days
Turkeys: 12 days
Not authorised for use in birds producing eggs for human consumption.

Packing: 500 g

Registration holder and manufactured by:

Rhone Ma Malaysia Sdn Bhd (200001023279)
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Date of revision of package insert:
5LRDOXY3-500 01/XXXXX

LABEL CODE: 5LRDOXY3-500	**VERSION**	**COLORS**
PRODUCT: RHODOXY 50WSP POWDER FOR ORAL SOLUTION 50%W/W PACK SIZE: 500 g COUNTRY: MY DIMENSIONS: cm = xxxx	Version: 1 MMM/YY:	

Marketing approval:	**Regulatory** approval:	**GMP** approval:
COMPLETE NAME: DATE: SIGNATURE:	COMPLETE NAME: DATE: SIGNATURE:	COMPLETE NAME: DATE: SIGNATURE: